Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/05/2024
PI Name:	Lawler, John Ph.D.		
Project Title:	Upstream Regulation of Nox2 and Skeletal Muscle Atrophy During Microgravity and Countermeasure Development		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	06/01/2019	**End Date:**	05/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2025 per NSSC information (Ed., 6/21/24). NOTE: End date changed to 05/31/2024 per NSSC information (Ed., 11/14/23). NOTE: End date changed to 05/31/2023 per F. Hernandez/ARC (Ed., 6/20/22). NOTE: Start/end dates changes to 6/1/2019-7/31/2022 (originally 3/1/2019-5/31/2022) per NSSC award documents per F. Hernandez/ARC (Ed., 12/3/2020)		
Key Personnel Changes/Previous PI:	August 2020 report: Dr. Khaled Kamal - Assistant Research Scientist ; Amin Mohajeri - PhD student; Mariam Othman - PhD student (add) Adding Joo Kim (PhD student) Jordyn Johnson - MS student. (remove - left because of COVID) Add: Aggie Research Scholars: Devon Roeming, Samhitha Harvey, Danielle DeCastro, Gracie Barrow, Yasmin Bagheri Victoria Frank, Anish Kumaran, Mickael Gergheis, Kyle Hawley, Fernanda Sifuentes (Aggie Research Scholar) 2022- Graduated: Myles McFarland (Aggie Research Scholar) 2020- ; Mollie Linder (Aggie Research Scholar) 2020- ; Hallie Harris (Aggie Research Scholar) 2020 - ; Francisco Melesio (Aggie Research Scholar) 2020 - ; Sonny Rodriguez (Aggie Research Scholar) 2020- ; Mia Ngyuen (Aggie Research Scholar) 2020- ; Lorrie Hill 2018-2020 (Kinesiology Research Scholar).		
COI Name (Institution):	Fluckey, James Ph.D. (Texas A & M, College Station)		

Grant/Contract No.:	80NSSC19K0432
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Skeletal muscles are dynamic mechanical and metabolic machines that drive body movement and energy expenditure. Skeletal muscles are dynamic tissues that can adapt quickly to alterations in mechanical unloading by altering their mass and muscle fiber cross-sectional area. Skeletal muscle strength and endurance are essential to the health, well-being, and performance of astronauts during spaceflight and upon return to a gravitational environment. The mechanical unloading due to the microgravity (µG) of spaceflight causes muscle fiber atrophy and fiber-type shift of postural muscles in the lower extremities and flexors in the upper extremities. Microgravity also increases the risk of skeletal muscle damage, weakness, and thus the risk of injury upon reloading (e.g., extravehicular activities, Mars). Mechanotransduction is the ability to sense and regulate adaptive responses to increased or decreased loading. New paradigms have emerged from ground analogs of µG that have contributed to a leap of our understanding of mechanotransduction and muscle atrophy. Specifically, the mu-splice variant of neuronal nitric oxide synthase (nNOSµ) was discovered as causal in both muscle hypertrophy with overloading and atrophy with unloading. Our laboratory has found that reactive oxygen species (ROS) directly contribute to both muscle fiber atrophy and fiber-type shift from slow to fast. Pilot Data and cutting-edge research have identified mitochondria, the Nox2 isoform of NADPH oxidase, and upstream angiotensin II receptor 1 (AT1R) as sources of ROS during mechanical unloading. Preliminary Data show that inhibition of Nox2 translocation of nNOSµ away from the sarcolemma, muscle fiber atrophy, and fiber-type shift. However, the upstream mechanisms that regulate Nox2 during µG are poorly understood, impeding progress in space biology and novel countermeasure development. The lack of such knowledge impedes our development in understanding the mechanisms that underlie redox regulation of mechanotransduction in skeletal muscle. This grant application serves as a renewal and extension and Renewal of our research team's NNX13AE45G award, particularly stretching our horizons in understanding how Nox2 assembly is enhanced during microgravity in skeletal muscle. New studies have identified novel inhibitors for proteins recently govern assembly of the Nox2 complex at the cell membrane acid—sphingomyelinase (ASMase) and cyclophilin A. We hypothesize that the following novel countermeasures will protect against nNOSµ translocation and the spaceflight phenotype— (a) the ASMase inhibitor etidronate (Didronel) and (b) cyclophilin A inhibitor TMN-355. We further postulate that Nox2 is causal in ROS-induced suppression of anabolic signaling. The efficacy and specificity of the above countermeasures will be confirmed with gene knockdown experiments. Texas A&M is a rich research environment for NASA research, including the Space Life Science Program. We will use the latest molecular and image analysis tools in the development of highly novel countermeasures against spaceflight sarcopenia during microgravity. Dr. Lawler and Dr. Fluckey's laboratories have continued to be supported by NASA, and are dedicated to finding targeted, antioxidant countermeasures against spaceflight sarcopenia. The ground hindlimb unloading model will be used in short and long-term experiments. Our research will also directly translate to skeletal muscle wasting in clinical setting on Earth, an important mission of the Space Biology program. For example, hospitalization, particularly in an ICU (intensive care unit) can reduce skeletal muscle by 25%. Cast immobilization can decrease affected muscle mass by 30% as well. In addition, mechanical unloading due to disuse (e.g., bedrest) and illness (e.g., sepsis, chronic obstructive pulmonary disease, chronic heart failure) exacerbates atrophy, weakness, and impedes recovery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will also directly translate to skeletal muscle wasting in clinical setting on Earth, an important mission of the Space Biology program. For example, hospitalization, particularly in an ICU (intensive care unit) can reduce skeletal muscle mass by 25%. Cast immobilization can decrease affected muscle mass by 30% as well. In addition, mechanical unloading due to disuse (e.g., bedrest) and illness (e.g., sepsis, chronic obstructive pulmonary disease, chronic heart failure) exacerbates atrophy, weakness, and impedes recovery.
Task Progress:	Hindlimb unloading in rodents will be conducted with an adaptation of the technique previously described by our laboratory (Hord 2021; Lawler 2021). We employed a hindlimb harness to produce unloading, as previously described by Mortreux (2020). The copper wire harness is overfitted with elastic tubing and moleskin as a pliable cushion. The harness will be hoisted using a stainless-steel chain attached to a fishing tackle swivel, which is allowed to move freely on a lubricated brass rod. The hindlimb harness will allow rodents to maneuver around the cage with their hindlimb paws just above the cage bedding. Young Adult Fischer 344 (4-5 months old) male rats were used for the following experimental groups (n=6 / group): 1- Group A: Ambulatory Controls, ‘Rats will be housed in the same HU cages.’ 2- Group B: hindlimb unloaded for 10 days (HU). 3- Group C: hindlimb unloading (10 days) + SRT2104 (HUS). 4- Group D: hindlimb unloading (10 days) + TRIM (HUT). A new group/experiment (Group D) was designed and performed to investigate the application of Time Release Ion Matrix (TRIM) to mitigate muscle atrophy, promote recovery, and reduce fibrosis in disuse-induced skeletal muscle atrophy. At the end of the experiment, tissues were dissected and snap-freezing for further histochemical and western blot analyses. In addition, the sirtuin-1 (SIRT1) agonist SRT2014 applied during 10 days of hindlimb unloading attenuated the unloading-induced elevation of RANK ligand (RANKL) [Wesolowski 2013]. Soleus and quadriceps atrophy were significantly protected by SRT2104. SIRT1 augmentation also protected against elevation of RANKL and the Nox2 subunit p67phox. Upstream regulators of Nox2 (e.g., cyclophilin A, acid sphingomyelinase) increased with unloading, which was prevented by SRT2104. Calpain activity, reflective of Ca2+ overload and catabolism, was more than doubled with unloading but mitigated by SRT2014. alpha-Klotho dimers were elevated with unloading but there appeared to be cleavage with SRT-2104. Elevated muscle levels of alpha-Klotho could lead to elevation of autophagy or suppression of Akt-mTOR signaling

	[deCavanagh 2015]. Furthermore, in situ muscle force protocol has been performed, using the Aurora system on the transversus abdominis (TA) muscle. Data reveals that the mechanical unloading reduces the TA muscle force. Moreover, the activation of Sirt1 using SRT2104 shows a partial protection of the muscle force in unloaded rats (Preliminary Data). Data shows that the upregulation of SIRT1 (using SRT2104) throughout the unloading period protects skeletal muscle localization of nNOS at the cell membranes.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
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